Complex Numbers: C + iS

Questions 1–43

Question 1. Identify whether each of the following series are a geometric series or a binomial series:

a) $1+\frac{1}{3}+\frac{1}{9}+\cdots$
b) $8-1.2+0.06-0.001$
c) $2-\frac{1}{2}+\frac{1}{8}-\cdots-\frac{1}{16384}$
d) $1+0.8+0.24+0.032+0.0016$

Question 2. Given that each of the following infinite series are convergent, find the sum of each giving your answer in the form $\frac{A}{B-e^{iC\theta}}$, where A, B and C are constants to be found.

a) $4+\frac{4e^{i\theta}}{3}+\frac{4e^{2i\theta}}{9}+\frac{4e^{3i\theta}}{27}+\cdots$
b) $1+\frac{e^{i\theta}}{4}+\frac{e^{2i\theta}}{16}+\frac{e^{3i\theta}}{64}+\cdots$
c) $2+\frac{2e^{2i\theta}}{5}+\frac{2e^{4i\theta}}{25}+\frac{2e^{6i\theta}}{125}+\cdots$
d) $5+\frac{e^{3i\theta}}{2}+\frac{e^{6i\theta}}{20}+\frac{e^{9i\theta}}{200}+\cdots$
e) $6+\frac{e^{-i\theta}}{2}+\frac{e^{-2i\theta}}{24}+\frac{e^{-3i\theta}}{288}+\cdots$
f) $2+\frac{e^{-4i\theta}}{3}+\frac{e^{-8i\theta}}{18}+\frac{e^{-12i\theta}}{108}+\cdots$

Question 3. Each of the following form a finite geometric series where the common ratio has a value not equal to 1. In each case, determine the sum:

a) $e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \cdots + e^{12i\theta}$

b) $2 + e^{i\theta} + \frac{e^{2i\theta}}{2} + \cdots + \frac{e^{9i\theta}}{256}$

c) $1 - \frac{e^{i\theta}}{3} + \frac{e^{2i\theta}}{9} - \cdots + \frac{e^{8i\theta}}{6561}$

d) $16 - 4e^{2i\theta} + e^{4i\theta} - \cdots - \frac{e^{14i\theta}}{1024}$

Question 4. State the complex conjugate of each of the following:

a) $z = a + ib$

b) $z = r(\cos\theta + i\sin\theta)$

c) $z = re^{i\theta}$

Question 5. State the complex conjugate of each of the following:

a) $z = 5 + e^{i\theta}$
b) $z = 3 - e^{-i\theta}$
c) $z = 7 + e^{2i\theta}$
d) $z = 7 + 9e^{i\theta}$
e) $z = 2 - 3e^{i\theta}$
f) $z = 1 + e^{-i\theta}$
g) $z = 2 - e^{i\theta}$
h) $z = 5 - 5e^{-i\theta}$
i) $z = 6 - 8e^{-7i\theta}$
j) $z = 2 + 15e^{i\theta/2}$

Question 6. Evaluate and simplify each of the following:

a) $(1 + 11e^{-i\theta})(1 + 11e^{i\theta})$
b) $(4 + e^{3i\theta})(4 + e^{-3i\theta})$
c) $\left(2 + \frac{1}{2}e^{i\theta}\right)\left(2 + \frac{1}{2}e^{-i\theta}\right)$

Question 7. It is given that $z = e^{i\theta}$.

a) Write z in terms of sine and cosine.

b) Show that $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$.

Question 8. Write each of the following in the form $a + b\cos(p\theta)$, where a, b and p are constants to be found:

a) $10 + 6e^{i\theta} + 6e^{-i\theta}$

b) $5 - 2e^{4i\theta} - 2e^{-4i\theta}$

c) $(1 + e^{3i\theta})(1 + e^{-3i\theta})$

d) $(2 + e^{2i\theta})(2 + e^{-2i\theta})$

e) $\left(1 - \frac{1}{2}e^{i\theta/3}\right)\left(1 - \frac{1}{2}e^{-i\theta/3}\right)$

Question 9. Rationalise the denominator for each of the following, in each case using $2\cos\theta = e^{i\theta} + e^{-i\theta}$ to write the denominator in terms of $\cos\theta$:

a) $\frac{1}{2-e^{i\theta}}$

b) $\frac{3}{4+e^{2i\theta}}$

c) $\frac{5}{6-2e^{-i\theta}}$

d) $\frac{e^{i\theta}}{1-\frac{1}{3}e^{3i\theta}}$

e) $\frac{e^{i\theta}}{1+\frac{1}{8}e^{4i\theta}}$

Question 10. Write $C + iS$ as a sum of three exponential terms, given that $C = \cos\theta + \frac{1}{4}\cos 3\theta + \frac{1}{16}\cos 5\theta$ and $S = \sin\theta + \frac{1}{4}\sin 3\theta + \frac{1}{16}\sin 5\theta$.

Question 11. Write $C + iS$ as a sum of three exponential terms, given that $C = 1 - \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos 4\theta$ and $S = -\frac{1}{2}\sin 2\theta + \frac{1}{4}\sin 4\theta$.

Question 12. Convergent infinite series C and S are defined by $C = \cos\theta - \frac{1}{3}\cos 4\theta + \frac{1}{9}\cos 7\theta - \cdots$ and $S = \sin\theta - \frac{1}{3}\sin 4\theta + \frac{1}{9}\sin 7\theta - \cdots$.

a) Expand and simplify $\left(1 + \frac{1}{3}e^{3i\theta}\right)\left(1 + \frac{1}{3}e^{-3i\theta}\right)$, writing your answer in the form $a + b\cos(3\theta)$, where a and b are constants to be found.

b) Find the sum of $C + iS$.

c) Show that $C = \frac{9\cos\theta + 3\cos 2\theta}{10 + 6\cos 3\theta}$.

d) Show that $S = \frac{9\sin\theta - 3\sin 2\theta}{10 + 6\cos 3\theta}$.

Question 13. Convergent infinite series C and S are defined by $C = \cos 3\theta + \frac{1}{2}\cos 5\theta + \frac{1}{4}\cos 7\theta + \cdots$ and $S = \sin 3\theta + \frac{1}{2}\sin 5\theta + \frac{1}{4}\sin 7\theta + \cdots$.

a) Show that $C = \frac{4\cos 3\theta - 2\cos\theta}{5 - 4\cos 2\theta}$.

b) Show that $S = \frac{4\sin 3\theta - 2\sin\theta}{5 - 4\cos 2\theta}$.

Question 14. Convergent infinite series C and S are defined by $C = 3 + \cos\theta + \frac{1}{3}\cos 2\theta + \cdots$ and $S = \sin\theta + \frac{1}{3}\sin 2\theta + \cdots$.

a) Show that $C = \frac{27 - 9\cos\theta}{10 - 6\cos\theta}$.

b) Show that $S = \frac{9\sin\theta}{10 - 6\cos\theta}$.

Question 15. Finite series C and S are defined by $C = 2 + \cos 2\theta + \frac{1}{2}\cos 4\theta + \cdots + \frac{1}{256}\cos 18\theta$ and $S = \sin 2\theta + \frac{1}{2}\sin 4\theta + \cdots + \frac{1}{256}\sin 18\theta$.

a) Show that $\left(1 - \frac{1}{2}e^{2i\theta}\right)\left(1 - \frac{1}{2}e^{-2i\theta}\right) = \frac{5}{4} - \cos 2\theta$.

b) Show that $C = \frac{2048 - 1024\cos 2\theta - 2\cos 20\theta + \cos 18\theta}{256(5 - 4\cos 2\theta)}$.

c) Show that $S = \frac{1024\sin 2\theta - 2\sin 20\theta + \sin 18\theta}{256(5 - 4\cos 2\theta)}$.

Question 16. Finite series C and S are defined by $C = \cos\theta + \cos 3\theta + \cos 5\theta + \cdots + \cos 45\theta$ and $S = \sin\theta + \sin 3\theta + \sin 5\theta + \cdots + \sin 45\theta$.

a) Show that $(1 - e^{2i\theta})(1 - e^{-2i\theta}) = 2 - 2\cos 2\theta$.

b) Show that $C = \frac{-\cos 47\theta + \cos 45\theta}{2 - 2\cos 2\theta}$.

c) Show that $S = \frac{2\sin\theta - \sin 47\theta + \sin 45\theta}{2 - 2\cos 2\theta}$.

Question 17.

a) Show that $\left(1-\frac{1}{9}e^{3i\theta}\right)\left(1-\frac{1}{9}e^{-3i\theta}\right)=\frac{1}{81}(82-18\cos 3\theta)$.

b) Find the sum of the convergent series $e^{3i\theta}+\frac{1}{9}e^{6i\theta}+\frac{1}{81}e^{9i\theta}+\cdots$, giving your answer in terms of $e^{i\theta}$.

c) Hence show that $C=\frac{81\cos 3\theta-9}{82-18\cos 3\theta}$.

d) Find a similar expression for $S=\sum_{n=1}^{\infty}\frac{1}{9^{n-1}}\sin(3n\theta)$.

Question 18. $C=\frac{1}{3}\cos\theta+\frac{1}{9}\cos 2\theta+\frac{1}{27}\cos 3\theta+\cdots+\frac{1}{3^n}\cos n\theta$ and $S=\frac{1}{3}\sin\theta+\frac{1}{9}\sin 2\theta+\frac{1}{27}\sin 3\theta+\cdots+\frac{1}{3^n}\sin n\theta$. Show that $C=\frac{3^{n+1}\cos\theta-3^n-3\cos((n+1)\theta)+\cos(n\theta)}{3^n(10-6\cos\theta)}$.

Question 19.

a) Show that $C = \frac{-\cos((2n+1)\theta)+\cos((2n-1)\theta)}{2-2\cos 2\theta}$.

b) Show that $S = \frac{2\sin\theta-\sin((2n+1)\theta)+\sin((2n-1)\theta)}{2-2\cos 2\theta}$.

Question 20. Use Binomial expansion to expand each of the following:

a) $(1+e^{i\theta})^4$

b) $(1+e^{i\theta})^5$

c) $(1+e^{i\theta})^n$

d) $(1+e^{2i\theta})^n$

Question 21.

a) Find $1 + \cos 2A$ in terms of $\cos A$.

b) State $\sin 2A$ in terms of $\sin A$ and $\cos A$.

c) Hence show that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

Question 22. Finite series C and S are defined by $C = 1 + 4\cos\theta + 6\cos 2\theta + 4\cos 3\theta + \cos 4\theta$ and $S = 4\sin\theta + 6\sin 2\theta + 4\sin 3\theta + \sin 4\theta$.

a) Show that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

b) Show that $C = 16\cos^4(\frac{1}{2}\theta)\cos(2\theta)$.

c) Hence, show that $S = 16\cos^4(\frac{1}{2}\theta)\sin(2\theta)$.

Question 23. Finite series C and S are defined by $C = 1 - 5\cos\theta + 10\cos 2\theta - \cdots - \cos 5\theta$ and $S = -5\sin\theta + 10\sin 2\theta - \cdots - \sin 5\theta$.

a) Show that $1 - e^{i\theta} = -2ie^{i\theta/2}\sin\frac{1}{2}\theta$.

b) Show that $C = 32\sin^5(\frac{1}{2}\theta)\sin(\frac{5}{2}\theta)$.

c) Hence, show that $S = -32\cos^5(\frac{1}{2}\theta)\cos(\frac{5}{2}\theta)$.

Question 24. Finite series C and S are defined by $C = 1 + 6\cos\theta + 15\cos 2\theta + \cdots + \cos 6\theta$ and $S = 6\sin\theta + 15\sin 2\theta + \cdots + \sin 6\theta$. It is known that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

a) Show that $C = 64\cos^6(\frac{1}{2}\theta)\cos(3\theta)$.

b) Hence, show that $S = 64\cos^6(\frac{1}{2}\theta)\sin(3\theta)$.

Question 25. Finite series C and S are defined by $C = 1 + 6\cos 2\theta + 15\cos 4\theta + \cdots + \cos 12\theta$ and $S = 6\sin 2\theta + 15\sin 4\theta + \cdots + \sin 12\theta$. It is known that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

a) Show that $C = 64\cos 6\theta\cos^6\theta$.

b) Hence, show that $S = 64\sin 6\theta\cos^6\theta$.

Question 26. Finite series C and S are defined by $C = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \cos n\theta$ and $S = \binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + \sin n\theta$. It is known that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

a) Show that $C = 2^n\cos^n(\frac{1}{2}\theta)\cos(\frac{n}{2}\theta)$.

b) Hence, show that $S = 2^n\cos^n(\frac{1}{2}\theta)\sin(\frac{n}{2}\theta)$.

Question 27. Finite series C and S are defined by $C = 1+3\cos\theta+3\cos 2\theta+\cos 3\theta$ and $S = 3\sin\theta+3\sin 2\theta+\sin 3\theta$.

a) Show that $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{1}{2}\theta)$.

b) Show that $C = 8\cos^3(\frac{1}{2}\theta)\cos(\frac{3}{2}\theta)$.

c) Hence, show that $S = 8\cos^3(\frac{1}{2}\theta)\sin(\frac{3}{2}\theta)$.

Question 28.

a) Given that $|z| < 1$, write down the sum of the infinite series $1 + z + z^2 + z^3 + \cdots$.

b) Hence, given that $z = \frac{1}{3}(\cos\theta + i\sin\theta)$, show that $\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots = \frac{3\sin\theta}{10-6\cos\theta}$.

c) Show that the sum cannot be purely imaginary, giving a reason for your answer.

Question 29.

a) Given that $|z| < 1$, write down the sum of the infinite series $z + z^2 + z^3 + \cdots$.

b) Hence, given that $z = \frac{1}{4}(\cos\theta + i\sin\theta)$, show that $\frac{1}{4}\cos\theta + \frac{1}{16}\cos 2\theta + \frac{1}{64}\cos 3\theta + \cdots = \frac{4\cos\theta - 1}{17 - 8\cos\theta}$.

c) Determine for which values of θ the sum is purely real. Give your answer in terms of π for any integer values.

Question 30. An Argand diagram shows points $z, z^2, z^3, \ldots$ where $z = \frac{1}{2}e^{i\theta}$.

a) What is the ratio of successive distances of the points from the origin?

b) What is the angle between successive position vectors on the Argand diagram?

c) Show that $\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots = \frac{2\sin\theta}{5 - 4\cos\theta}$.

Question 31. Series C and S are defined by $C = \frac{1}{2}\cos\theta + \frac{1}{4}\cos 2\theta + \frac{1}{8}\cos 3\theta + \cdots + \frac{1}{2^n}\cos n\theta$ and $S = \frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots + \frac{1}{2^n}\sin n\theta$. Show that $S = \frac{2^{n+1}\sin\theta - 2\sin((n+1)\theta) + \sin(n\theta)}{2^n(5-4\cos\theta)}$.

Question 32. Given $S = \sum_{r=0}^{12} \binom{12}{r} \sin(r\theta)$, show that $S = 4096\sin(6\theta)\cos^{12}(\frac{1}{2}\theta)$.

Question 33.

$$C = 1 - \binom{16}{1}\cos\theta + \cdots + \cos 16\theta, \quad S = 1 - \binom{16}{1}\sin\theta + \cdots + \sin 16\theta.$$

Show that $C = 2^{16}\cos(8\theta)\sin^{16}(\frac{1}{2}\theta)$.

Question 34.

a) Show that $\frac{2}{1-e^{i2\theta}} = 1 + i\cot\theta$.

b) Finite series C and S are defined by $C = 2+2\cos\frac{\pi}{14}+2\cos\frac{2\pi}{14}+\cdots+2\cos\frac{6\pi}{14}$ and $S = 2(\sin\frac{\pi}{14}+\sin\frac{2\pi}{14}+\cdots+\sin\frac{6\pi}{14})$. Show that $C = 1 + \cot\frac{\pi}{28}$.

c) Hence show that $S = C - 2$.

Question 35. You are given that $C = \cos\theta + \cos(\theta + \frac{4\pi}{5}) + \cos(\theta + \frac{8\pi}{5}) + \cos(\theta + \frac{12\pi}{5}) + \cos(\theta + \frac{16\pi}{5})$ and $S = \sin\theta + \sin(\theta + \frac{4\pi}{5}) + \sin(\theta + \frac{8\pi}{5}) + \sin(\theta + \frac{12\pi}{5}) + \sin(\theta + \frac{16\pi}{5})$.

a) Show that $C + iS = e^{i\theta}\sum_{k=0}^{4}(e^{4\pi i/5})^k$.

b) Show that $C = 0$ and $S = 0$.

Question 36. Let $C = \cos 0 - \cos\frac{\pi}{2} + \cos\pi - \cos\frac{3\pi}{2}$ and $S = \sin 0 - \sin\frac{\pi}{2} + \sin\pi - \sin\frac{3\pi}{2}$.

a) Write $C + iS$ in summation notation.

b) Show that $C = 0$ and $S = 0$.

Question 37. The infinite series C and S are defined as $C = 1 + k\cos\theta + k^2\cos 2\theta + k^3\cos 3\theta + \cdots$ and $S = k\sin\theta + k^2\sin 2\theta + k^3\sin 3\theta + \cdots$, where k is a real number and $|k| < 1$.

a) Show that $C = \frac{1-k\cos\theta}{1-2k\cos\theta+k^2}$.

b) Find a corresponding expression for S.

Question 38. The infinite series C and S are defined as $C = k\cos\theta + k^2\cos 2\theta + k^3\cos 3\theta + \cdots$ and $S = k\sin\theta + k^2\sin 2\theta + k^3\sin 3\theta + \cdots$, where k is a real number and $|k| < 1$.

a) Show that $C = \frac{k\cos\theta-k^2}{1-2k\cos\theta+k^2}$.

b) Find a corresponding expression for S.

Question 39. Finite series C and S are defined by $C = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \binom{n}{n}\cos n\theta$ and $S = \binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + \binom{n}{n}\sin n\theta$.

a) Show that $C = 2^n \cos^n(\frac{\theta}{2})\cos(\frac{n\theta}{2})$.

b) Find a corresponding expression for S.

c) Hence, show that $\frac{C}{S} = \cot(\frac{n\theta}{2})$.

Question 40. Finite series C and S are defined by $C = 1 - \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta - \cdots + (-1)^n \cos n\theta$ and $S = -\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta - \cdots + (-1)^n \sin n\theta$.

a) Show that $1 - e^{i\theta} = -2ie^{i\theta/2}\sin\frac{\theta}{2}$.

b) Show that $C + iS = \left[-2i\sin\frac{\theta}{2}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})\right]^n$.

c) Hence, show that, for even values of n, $\frac{S}{C} = \tan(\frac{n\theta}{2})$.

Question 41. It is given that $\int e^{(a+bi)x}\,dx = \frac{1}{a+bi}e^{(a+bi)x} + c$ for real values a and b. The integrals C and S are defined by $C = \int_0^{\pi/3} e^{3x}\cos 5x\,dx$ and $S = \int_0^{\pi/3} e^{3x}\sin 5x\,dx$.

a) By considering $C + iS$, show that $C = \frac{e^{\pi}}{68}(3 - 5\sqrt{3}) - \frac{3}{34}$.

b) Obtain a similar expression for S.

Question 42. The finite series C and S are defined as $C = 1 + \cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta$ and $S = \sin\theta + \sin 2\theta + \sin 3\theta + \sin 4\theta$.

a) Show that $C + iS = \frac{e^{5i\theta/2} - e^{-5i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}} \times e^{2i\theta}$.

b) Hence show that $C = \frac{1}{2}\sin(4\theta)\cot(\frac{1}{2}\theta) + \cos^2(2\theta)$.

c) Determine the values of θ in the range $0 < \theta < \pi$ for which $C = S$, giving your answer in exact form.

Question 43. It is given that $P = e^{i\theta} + e^{2i\theta} + e^{3i\theta} + e^{4i\theta}$.

a) State the value of P for $\theta = 2n\pi$, where n is an integer.

b) Show that for $\theta \neq 2n\pi$ for integer n, $P = \frac{e^{i\theta/2}(e^{4i\theta}-1)}{2i\sin(\frac{1}{2}\theta)}$.

c) Show that for $\theta \neq 2n\pi$ for integer n, $\cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta = \frac{\sin(\frac{9}{2}\theta)}{2\sin(\frac{1}{2}\theta)} - \frac{1}{2}$.

d) Hence show that $\theta = \frac{\pi}{5}$ is a root of $\cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta = 0$ and find another root in the interval $0 < \theta \leq \frac{\pi}{2}$.